

Project Workflow: Week 3

Phase: Environment Setup & Project Initialization (Bounce Ball Game)

1. Objective

The primary goal of Week 3 was to establish the complete Java development environment and initialize the Bounce Ball Game project structure. This included:

- Installing and configuring the Java Development Kit (JDK).
- Setting up the selected Integrated Development Environment (IDE).
- Configuring the Java Swing and AWT environment for graphical user interface development.
- Creating the project structure and main game window class.

2. Detailed Task Breakdown

JDK Installation & IDE Configuration

Installed the required Java Development Kit (JDK 11+) and configured the local development environment. The IDE was prepared for Java development, including project SDK verification and initial program execution testing.

Java Swing Environment Setup

Configured Java Swing and AWT libraries for building the game window and visual elements. Primary GUI components such as JFrame and JPanel.

Project Structure & Main Window Class Creation

Created the initial modular project structure and package organization. Built the main game window class (BounceBallGame) extending JFrame, setting up basic window properties including window title, fixed dimensions, close operation, and desktop centering.

3. Deliverables

- **Configured Environment:** JDK 11+ and IDE successfully installed and verified.
- **Swing/AWT Setup:** Java Swing environment fully configured for GUI development.
- **Project Structure:** Clean and organized Java project directory structure created.
- **Main Window Class:** Functional BounceBallGame JFrame class created and initialized.
- **Environment Test Execution:** Initial Java Swing test program executed without errors.

4. Tools & Resources

Development Environment

- Java JDK 11+

- IntelliJ IDEA / VS Code / Eclipse

Libraries & Frameworks

- Java Swing (javax.swing)
- Java AWT (java.awt)
- JFrame & JPanel